

# Exploring and cleaning data with R

## Lesson 2

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**Open R studio**

# By the end of today

- View and Understand data
- Selecting Columns
- Filtering Rows
- Mutating/Adding Columns
- Grouping and Summarizing Data
- Plotting in R

# Recap

 Load the packages  
 `library(data.table)`

 `library(data.table)`

 `require(data.table)`

 `male <- c(20, 30, 40)`

 `female <- c(40, 30, 20)`

 `month <- c("Jan", "Feb")`

 `data.frame(male, female, month)`

 `data_Africa <- read_excel[path = "../inputs/African_meningococci.xlsx"]`

 `data_Africa <- read_excel(path = ../inputs/African_meningococci.xlsx)`

# View and Understanding your data

```
#=====
# 1. to know the structure of your data
str(data_01_Africa)

# OR

glimpse(data_01_Africa)

#=====
# 2. view the column names
names(data_01_Africa)

# OR

colnames(data_01_Africa)

#=====
# 3. tidy up the column names.
data_02_Africa <- clean_names(data_01_Africa)

# confirm that the names of the columns have changed
names(data_01_Africa)
names(data_02_Africa)
```

```
#=====
# 4. To view the first top rows - by default will view 6
data_03_Africa <- head(data_01_Africa)

#=====
# 5. To view the bottom rows - by default will view the last 6
data_04_Africa <- tail(data_01_Africa)

#=====
# 6. Selecting columns
data_01_subset <- select(data_01_Africa, id, isolate, country)

# OR (remove certain columns)

data_02_subset <- select(data_01_Africa, -id, -isolate)

#=====
# 7. Filtering rows

Nigeria <- filter(data_01_Africa, country == "Nigeria")

#=====
# 8. Number of samples in each country
table(data_01_Africa$country)
```

# Questions we can answer?

1. The number of samples in each year? [hint: table]
2. The number of samples in each serogroup?
3. The number of samples collected in each year only in Togo?  
[hint: filter then table]

# Answers

1. The number of samples in each year? [hint: table]

| 2011 | 2012 | 2013 | 2014 | 2015 | 2016 |
|------|------|------|------|------|------|
| 115  | 272  | 60   | 26   | 158  | 85   |

2. The number of samples in each serogroup?

| A  | C   | NG | W   | X  | Y |
|----|-----|----|-----|----|---|
| 90 | 124 | 8  | 431 | 61 | 2 |

3. The number of samples collected in each year only in Togo?  
[hint: filter then table]

| 2014 | 2015 | 2016 |
|------|------|------|
| 16   | 12   | 42   |

# Too many variables

- Pipe operator: %>%
- Can I clean col names, select, filter all in one variable and run once.

```
#=====#  
# remove all the data frames loaded except the data_01_Africa  
rm(data_02_Africa, data_03_Africa, data_04_Africa, Nigeria, data_01_subset, data_02_subset)
```

# 9. clean names

```
data_02_Africa <- data_01_Africa %>%  
  clean_names()
```

# 9. clean names | selecting columns

```
data_02_Africa <- data_01_Africa %>%  
  clean_names() %>%  
  select(id, isolate, year, country)
```

# 9. clean names | selecting columns | Filter only "Burkina Faso"

```
data_02_Africa <- data_01_Africa %>%  
  clean_names() %>%  
  select(id, isolate, year, country) %>%  
  filter(country == "Burkina Faso")
```



# **Data Wrangling and Visualization**

## **Lesson 3**

**Leonard N., Dam K., Noutin M**

**18<sup>th</sup> - 19<sup>th</sup> September 2024**

# Bonus Task

- For all the entries with Togo in the column country, replace with TOGO. i.e. Togo -> TOGO

```
library(stringi)
library(stringr)

bonus_task <- data_01_Africa %>%
  clean_names() %>%
  mutate(country = str_replace(string = country, pattern = "Togo", replacement = "TOGO"))
```

| isolate       | country |
|---------------|---------|
| All           | togo    |
| Togo_00073_16 | TOGO    |
| Togo_00075_16 | TOGO    |
| Togo_M0800_16 | TOGO    |
| Togo_M0842_16 | TOGO    |
| Togo_00048_16 | TOGO    |
| Togo_00049_16 | TOGO    |

- Change the isolate name from Togo\_ to TOGO\_

# Bonus Task

```
bonus_task <- data_01_Africa %>%  
  clean_names() %>%  
  mutate(country = str_replace(string = country, pattern = "Togo", replacement = "TOGO")) %>%  
  mutate(isolate = str_replace(string = isolate, pattern = "Togo", replacement = "TOGO"))
```

OR

```
bonus_task <- data_01_Africa %>%  
  clean_names() %>%  
  mutate(country = str_replace(string = country, pattern = "Togo", replacement = "TOGO"),  
         isolate = str_replace(string = isolate, pattern = "Togo", replacement = "TOGO"))
```

# The number of samples per year in each country

| country ▲                | year ▲ | count ▲ |
|--------------------------|--------|---------|
| Benin                    | 2012   | 41      |
| Burkina Faso             | 2012   | 167     |
| Burkina Faso             | 2011   | 41      |
| Burkina Faso             | 2013   | 20      |
| Burkina Faso             | 2015   | 11      |
| Burkina Faso             | 2016   | 5       |
| Burkina Faso             | 2014   | 4       |
| Cameroon                 | 2012   | 4       |
| Central African Republic | 2016   | 23      |
| Central African Republic | 2015   | 7       |

# Grouping and summarizing data

- **group\_by** and **summarise** function

```
#=====
# 10. Number of samples identified in each year per Country
data_03_Africa <- data_01_Africa %>%
  clean_names() %>%
  group_by(country, year) %>%
  summarise(n())
```

```
# 10. Number of samples identified in each year per Country
data_03_Africa <- data_01_Africa %>%
  clean_names() %>%
  group_by(country, year) %>%
  summarise(count = n())
```

# More questions we can answer

1. The clonal complexes identified in each genogroup? [hint: group\_by and summarise]
2. The number of clonal complexes and genogroup identified by country? [hint: group\_by and summarise]

# Include the proportions of each



| country      | clonal_complex_mlst | count | prop       |
|--------------|---------------------|-------|------------|
| Benin        | ST-11 complex       | 38    | 92.6829268 |
| Benin        | ST-181 complex      | 3     | 7.3170732  |
| Burkina Faso | ST-10217 complex    | 1     | 0.4032258  |
| Burkina Faso | ST-11 complex       | 194   | 78.2258065 |
| Burkina Faso | ST-175 complex      | 2     | 0.8064516  |
| Burkina Faso | ST-181 complex      | 48    | 19.3548387 |

# 11. Number of samples identified in each year per Country

```
data_04_Africa <- data_01_Africa %>%  
  clean_names() %>%  
  group_by(country, clonal_complex_mlst) %>%  
  summarise(count = n()) %>%  
  mutate(prop = count/sum(count) *100)
```

# Task

1. Which Country has the highest proportion of Serogroup W?
2. Create one table showing the number of each serogroup by year and country\*\*\*\*



# The number of each serogroup by year and country

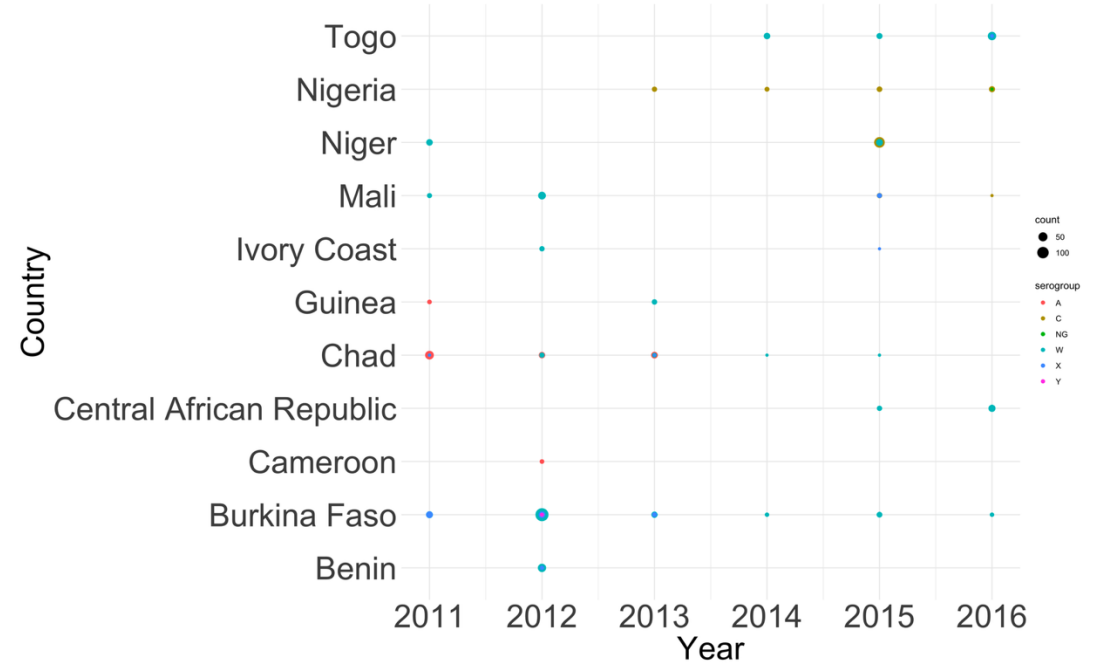
| country      | year | serogroup | count | prop      |
|--------------|------|-----------|-------|-----------|
| All          | All  | All       | All   | All       |
| Benin        | 2012 | W         | 38    | 92.682927 |
| Benin        | 2012 | X         | 3     | 7.317073  |
| Burkina Faso | 2011 | A         | 1     | 2.439024  |
| Burkina Faso | 2011 | W         | 21    | 51.219512 |
| Burkina Faso | 2011 | X         | 19    | 46.341463 |
| Burkina Faso | 2012 | NG        | 3     | 1.796407  |
| Burkina Faso | 2012 | W         | 140   | 83.832335 |
| Burkina Faso | 2012 | X         | 22    | 13.173653 |
| Burkina Faso | 2012 | Y         | 2     | 1.197605  |
| Burkina Faso | 2013 | W         | 14    | 70.000000 |

```
# 12. The number of each serogroup by year and country
data_05_Africa <- data_01_Africa %>%
  clean_names() %>%
  group_by(country, year, serogroup) %>%
  summarise(count = n()) %>%
  mutate(prop = count/sum(count) *100)
```

# Function: ggplot

| country                  | year | serogroup | count | prop       |
|--------------------------|------|-----------|-------|------------|
| Benin                    | 2012 | W         | 38    | 92.682927  |
| Beni NG                  | 2012 | X         | 3     | 7.317073   |
| Burkina Faso             | 2011 | A         | 1     | 2.439024   |
| Burkina Faso             | 2011 | W         | 21    | 51.219512  |
| Burkina Faso             | 2011 | X         | 19    | 46.341463  |
| Burkina Faso             | 2012 | NG        | 3     | 1.796407   |
| Burkina Faso             | 2012 | W         | 140   | 83.832335  |
| Burkina Faso             | 2012 | X         | 22    | 13.173653  |
| Burkina Faso             | 2012 | Y         | 2     | 1.197605   |
| Burkina Faso             | 2013 | W         | 14    | 70.000000  |
| Burkina Faso             | 2013 | X         | 6     | 30.000000  |
| Burkina Faso             | 2014 | NG        | 1     | 25.000000  |
| Burkina Faso             | 2014 | W         | 3     | 75.000000  |
| Burkina Faso             | 2015 | NG        | 1     | 9.090909   |
| Burkina Faso             | 2015 | W         | 10    | 90.909091  |
| Burkina Faso             | 2016 | NG        | 2     | 40.000000  |
| Burkina Faso             | 2016 | W         | 3     | 60.000000  |
| Cameroon                 | 2012 | A         | 4     | 100.000000 |
| Central African Republic | 2015 | W         | 7     | 100.000000 |
| Central African Republic | 2016 | W         | 23    | 100.000000 |
| Chad                     | 2011 | A         | 45    | 95.744681  |
| Chad                     | 2011 | W         | 1     | 2.127660   |
| Chad                     | 2011 | X         | 1     | 2.127660   |
| Chad                     | 2012 | A         | 16    | 69.565217  |
| Chad                     | 2012 | W         | 7     | 30.434783  |
| Chad                     | 2013 | A         | 19    | 79.166667  |
| Chad                     | 2013 | W         | 4     | 16.666667  |
| Chad                     | 2013 | X         | 1     | 4.166667   |
| Chad                     | 2014 | W         | 1     | 100.000000 |
| Chad                     | 2015 | W         | 1     | 100.000000 |
| Guinea                   | 2011 | A         | 4     | 100.000000 |
| Guinea                   | 2013 | A         | 1     | 11.111111  |
| Guinea                   | 2013 | W         | 8     | 88.888889  |
| Ivory Coast              | 2012 | W         | 7     | 100.000000 |
| Ivory Coast              | 2015 | X         | 1     | 100.000000 |
| Mali                     | 2011 | W         | 6     | 100.000000 |
| Mali                     | 2012 | W         | 30    | 100.000000 |
| Mali                     | 2015 | C         | 8     | 53.333333  |

OR



# Plotting the number of Serogroups in each Country by Year

```
# 13. Visualize with points on a graph
# We use the function ggplot, which is under ggplot2 package
library(ggplot2)
plot_Africa <- data_05_Africa %>%
  ggplot(aes(x = year, y = country)) +
  geom_point(aes(color = serogroup, size = count), alpha = 6)
```

```
# 13. Visualize with points on a graph
# We use the function ggplot, which is under ggplot2 package
library(ggplot2)
plot_Africa <- data_05_Africa %>%
  ggplot(aes(x = year, y = country)) +
  geom_point(aes(color = serogroup, size = count), alpha = 6) +
  xlab("Year") +
  ylab("Country") +
  theme_minimal() +
  theme(axis.title = element_text(size = 35),
        axis.text = element_text(size = 35),
        legend.text = element_text(size = 14),      # Increase legend text size
        legend.title = element_text(size = 16),     # Increase legend title size
        legend.key.size = unit(1.5, "lines"))
```

# Task

- Plot the Serogroup C distribution by Country for each Year

# Useful Resources

- This book contains all you need: from installation of both R and Rstudio, basics analysis and complex statistical analysis

<https://ggnindia.dronacharya.info/Downloads/Sub-info/RelatedBook/8thSem/R-PROGRAMMING-text-book-2.pdf>

**Thank you**